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Paleomagnetism of the Dockum Group (Upper Triassic), northwest Texas: Further evidence for the J-1 cusp in the North America apparent polar wander path and implications for rate of Triassic apparent polar wander and Colorado plateau rotation

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Abstract. We report paleomagnetic data for 26 accepted sites collected in two sections of flat-lying strata of the upper Carnian-lower Norian (~225 Ma) Dockum Group, northwest Texas. Six additional sites in coarse-grained conglomeratic sandstones gave no usable results. The total assemblage of 26 VGPs is streaked along the Late Triassic - earliest Jurassic track of the North America apparent polar wander path and their mean is inconsistent with the accepted upper Carnian-lower Norian reference pole. In detail, 12 sites in grayish white (nonhematitic) sandstones have weak magnetizations (less than about 1 mA/m) carried by magnetite or maghemite that give a paleopole at 56.4°N-96.3°E (N=12 dual-polarity VGPs; K=44.2; A95=6.6°) in close agreement with other results for upper Carnian-lower Norian rocks in North America. The 14 remaining sites in tan and red-colored (hematitic) sandstones, siltstones, and claystones give high unblocking temperature characteristic magnetizations carried by hematite, with paleopoles at 59.0°N-53.8°E (normal; N=7 VGPs, K=62.4, A95=7.6°) and 59.3°N-77.8°E (reverse; N=7 VGPs, K=204.2, A95=4.2°). These poles fall along the younger track of poles and near the J-1 cusp of the North American APWP as defined by (unrotated) poles derived from Colorado plateau rocks. We suggest that the characteristic magnetization of the non-hematitic sandstones is an "early" magnetization, acquired during or soon after deposition. However, the characteristic magnetization of the red bed sites is interpreted as a secondary magnetization, for which we infer an earliest Jurassic age. The secondary origin for this magnetization is supported by the observation of conflicting magnetostratigraphies. The extreme westward position of the poles derived from red bed sites, particularly those with normal polarity, confirms the general position of the J-1 cusp indicated by poles in the Piedmont province, the Newark basin, and the Colorado plateau; it also suggests that the magnitude of rotation of the Colorado plateau is no greater than about 5°. We compile an apparent polar wander path for North America, including Colorado plateau data, which suggests a fast rate of apparent polar wander throughout the Triassic period (about 0.8°/m.y.) with a gradual increase that preceded the opening of the Atlantic.

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Introduction, Rotation of the Colorado Plateau

Between Late Permian and Late Triassic time the paleomagnetic pole of North America tracked westward from a well-defined position at about 50°N-120°E to a more controversial position at about 60°N-70°E (Figure 1a). In Early Jurassic time, rapid changes in plate motion associated with the breakup of Pangea resulted in a rapid change in the rate and direction of apparent polar wander, with the pole tracking eastward from Early Jurassic time onward [*Gordon et al.*, 1984; *May and Butler*, 1986]. What has been interpreted to be a sharp cusp (the J-1 cusp) in the apparent polar wander path (APWP) defines the reversal in the direction of plate motion. The J-1 cusp was predicted by plate motion models and later confirmed by paleomagnetic data for the earliest Jurassic (Hettangian-Sinemurian) Moenave Formation in Utah and Arizona [*Ekstrand and Butler*, 1989]. The J-1 cusp is roughly dated at ~200 Ma.

The Moenave pole, at about 58°N-52°E, is well defined but its usefulness in determining the location of the J-1 cusp is controversial. Central to the debate is the fact that the Moenave Formation lies on the Colorado plateau, which is argued to have experienced a small, though geologically appreciable, clockwise rotation with respect to the craton [*Hamilton*, 1981; *Steiner*, 1988; *Bryan and Gordon*, 1990]. Rotation of the Colorado plateau affects other poles used to define the North American APWP. In fact, Late Carboniferous through Early Jurassic poles from the Colorado plateau (Figure 1a, solid symbols) fall consistently to

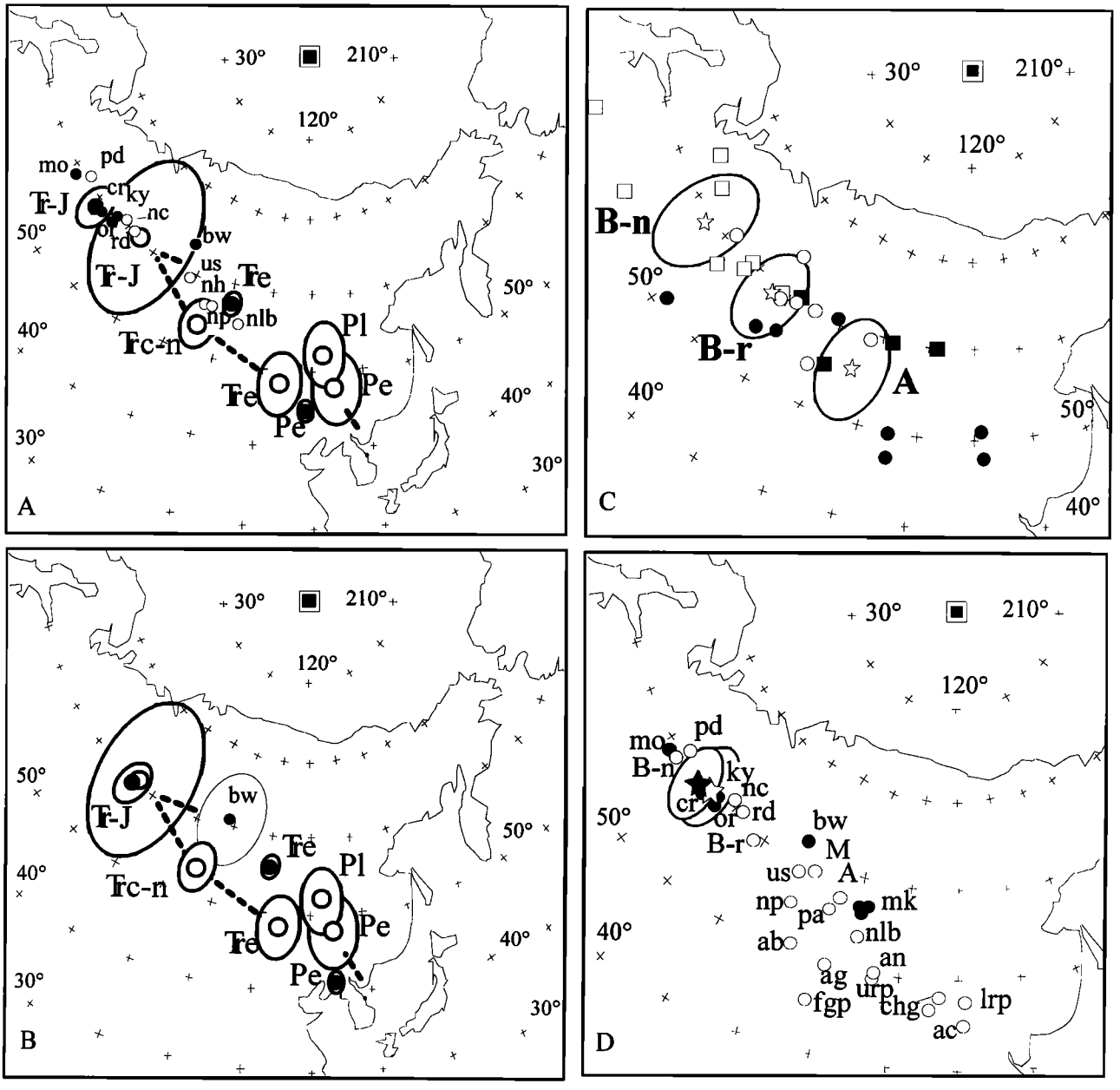


Figure 1. (a) Early Mesozoic North American apparent polar wander path showing the discrepancy between cratonic (large open symbols with 95% confidence regions) and Colorado plateau poles (large solid symbols); the distribution of poles has been interpreted to reflect rotation of the plateau. (Abbreviations are Pe, Early Permian, 295-265 Ma; Pl, Late Permian, 265-250 Ma; Tre,Early Triassic, 250-230 Ma; Trc-n, Late Triassic, 230-215 Ma; Tr-J, Triassic-Jurassic, 215-190 Ma). Small symbols are used for individual paleomagnetic results, with solid symbols for plateau poles. Note the large separation between paleomagnetic data from the Newark Basin (nlb, np, and nh) and the Chinle Group (bw, cr, or, us, rd). A key to the poles is listed in Table 1. (b) Same as Figure 1a with mean poles for the Colorado plateau rotated 5° about an Euler pole at 36°N-105°W. (c) Site VGPs from this study and mean Dockum Group poles A, B-n, and B-r (open stars with confidence regions). Open (solid) symbols are used for sites where the ChRM is dominantly carried by hematite (magnetite); squares (circles) are used for sites collected at the upper Palo Duro (lowe Palo Duro) locality. (d) Triassic and Early Jurassic North American poles used in constructing Figure 7. Closed symbols are unrotated poles from strata on the Colorado plateau. Pole symbols are the same as Table 1. The open and closed stars are cratonic and Colorado plateau mean poles (215-190 Ma), respectively, with their associated confidence regions.

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the west of "cratonic" poles (Figure 1a, open symbols), that is, poles obtained from sites assumed to have remained stable relative to the craton [*Bryan and Gordon*, 1990; *Steiner*, 1988]. There is little doubt that the Colorado plateau has experienced some clockwise rotation, but the amount of rotation is hotly debated.

The two principal sources of Late Triassic-Early Jurassic poles are nonmarine Triassic and Jurassic rocks of southwest North America and rift deposits of the Newark Supergroup on the Atlantic seaboard (Table 1). These data sets have been interpreted to indicate different amounts and rates of APW, further complicating the controversy over the plateau rotation [*Bazard and Butler*, 1991; *Kent and Witte*, 1993; *Witte and Kent*, 1990; *Molina-Garza* et al., 1991; *Kodama et al.*, 1994b]. Combined with paleomagnetic results for the Moenave Formation, the Chinle Group of southwest North America records ~25 million years of APW, a period that spans the late Carnian (~225 Ma) through Hettangian (~200 Ma). Individual poles within stratigraphic succession both on and off the plateau (Figure 1a) are spread over a ~15° angular distance [*Bazard and Butler*, 1991; *Molina-Garza et al.*, 1991]. These data suggest a fast rate of APW (greater than ~0.7°/m.y.).

In contrast, paleomagnetic data for the Newark Supergroup have been interpreted to indicate a small amount and thus a slow rate of APW. The angular distance between late Carnian and Hettangian poles derived from rocks in the Newark basin (poles nlb and nh, Figure 1a) is only ~5°, giving a rate of polar wander of 0.2°/m.y. [*Witte et al.*, 1991; *Kent and Witte*, 1993]. On the basis of the lack of evidence for a sharp J-1 cusp in the Newark basin, *Kent and Witte* [1993] have argued that the extreme west longitude for the Moenave pole is an artifact that resulted from a large (>10°) Colorado plateau rotation. Early Jurassic rocks of the extrusive zone of the Newark Supergroup [*Witte and Kent*, 1990] give a paleomagnetic pole separated by 23.1° +/- 5.7° from the (coeval) Moenave Formation pole. To bring these poles into agreement requires a clockwise rotation of the Colorado plateau of ~15°. Such a large rotation is, however, inconsistent with available geologic evidence [*Hamilton*, 1988; *Chase et al.*, 1992; *Karlstrom and Daniel*, 1993].

The location of some poles from rocks on the craton weakens the interpretation of data from the Newark basin. The poles for the Redonda Formation [*Reeve and Helsely*, 1972], the Culpeper basin [*Kodama et al.*, 1994b], and the Piedmont dikes [*Smith*, 1987] fall to the west of the Newark Hettangian pole and along the trend of the Chinle poles on the plateau (Figure 1a). *Kodama et al.* [1994a, 1994b] have suggested that rocks in the Newark basin, Pennsylvania, and the Culpeper basin, Virginia, do in fact record a faster rate of APW than that suggested by *Kent and Witte* [1993]. The circa 200 Ma pole (nc) of *Kodama et al.* [1994b] supports the existence of a "sharp" J-1 cusp. After close examination of structural and paleomagnetic data, *Kodama et al.* [1994a] have demonstrated the complexities associated with the structural history of rift basins in northeast North America and the intricacy of the paleomagnetic record of rocks deposited there. A more modest correction of plateau poles than that proposed by *Kent and Witte* [1993] or *Steiner* [1988], which rotates plateau poles only 5° about an Euler pole at 36°N-105°W, brings the mean Early Permian through Early Jurassic poles from the plateau into close agreement with its cratonic counterparts (Figure 1b; Table 1c). A small rotation (~5°) has been supported

Table 1a. Mean Late Paleozoic and Triassic Reference Paleomagnetic Poles for the North America Craton

Pole	Location	*N*	A95, deg
Tr-J	60.7°N 73.7°E	4	10.1
Tr c-n	53.1°N 95.0°E	8	4.2
Tr e	47.0°N 113.9°E	6	4.7
P l	52.5°N 123.5°E	3	4.9
P e	47.3°N 122.3°E	5	5.7

Pole symbols are used in figures 1 and 7. Data are from *Van der Voo* [1990], Bazard and Butler [1991], *Pan and Symons* [1993], *Steiner and Lucas* [1992], *Kent and Witte* [1993], and *Kodama et al.* [1994b].

Table 1b. Units Providing Individual Paleomagnetic Poles for the North America Craton

Pole (symbol)	lat.°N, long.°E
Triassic-Jurassic (215-190 Ma)	
Piedmont dikes (pd)	55,62
Newark extrusive zone (nh)	55,94
Redonda Formation (rd)	61,72
Newark-Culpeper (nc)	61,69
Late Triassic (230-215 Ma)	
Chinle, upper shale (us)	57,88
Manicouagan Impact (M)	59,90
Newark Passaic Fm. (np)	56,95 (54,88)
Newark lower beds (nlb)	54,102
Popo Agie Fm. (pa)	56,96
Fundy Group (fgp)	45,97
Agamenticus Pluton (ag)	48,99
Abbot Pluton (ab)	48,92
Early Triassic (250-230 Ma)	
Ankareh Fm. (an)	51,105
Upper Red Peak (urp)	49,105
Lower Red Peak (lrp)	46,121
Chugwater Fm. (chg)	47,114
Chugwater Fm. (chg)	43,115
Anton Chico Fm., NM (ac)	43,121
Late Permian (265-250 Ma)	
Ochoan red beds	55,119
Guadalupian red beds	51,125
Dewey Lake Fm.	51,126
Early Permian (295-265 Ma)	
Leonardian red beds	52,119
Ingleside Fm.	46,122
Casper Fm.	51,113
Pictou red beds	42,127
Cap aux Meules	45,129

Symbols are used in Figs. 1 and 7

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Table 1c. Mean Late Paleozoic and Triassic Reference Paleomagnetic Poles for the Colorado Plateau

Pole	Location	[rotated]	*N*	A95, deg
Tr-J	57.9°N 62.1°E	58.5°N 71.5°E	4	4.3
Tr c-n	63.4°N 86.8°E	62.2°N 97.3°E	1	6.8
Tr e	57.5°N 100.5°E	55.5°N 108.9°E	6	1.7
P l				
P e	44.4°N 118.3°E	41.5°N 123.9°E	6	2.2

Colorado plateau poles are listed in situ and rotated 5° about an Euler pole at 36°N-105°W. Symbols are used in Figures 1 and 7. See Table 1a for data sources.

Table 1d. Units Providing Individual Paleomagnetic Poles for the Colorado Plateau

Pole (symbol)	lat.°N, long.°E
Triassic-Jurassic (215-190)	
Kayenta Fm. (ky)	59,67
Moenave Fm. (mo)	58,52
Chinle, Church Rock Mbr. (cr)	58,63
Chinle, Owl Rock Mbr. (or)	57,66
Late Triassic (230-215)	
Bluewater Creek Fm. (bw)	63,87
Early Triassic (250-230)	
Upper Moenkopi Fm. Colorado (mk)	55,103
Lower Moenkopi Fm., Colorado	57,97
Upper Moenkopi Fm. Arizona	58,101
Upper Moenkopi Fm. Colorado	57,100
Moenkopi Fm. New Mexico	58,100
Upper Maroon Fm., Colorado	60,102
Early Permian (295-265)	
Minturn Fm. and Maroon Fm.	43,115
Cutler Fm., Colorado	43,122
Cutler Fm., Utah	44,116
Cutler Fm. Lisbon Valley	44,118
Fountain Fm. and Lykins Fm.	48,119
Elephant Canyon Fm.	44,120

Symbols are used in Figs. 1 and 7

by *Bryan and Gordon* [1990], Bazard and Butler [1991], and *Molina-Garza et al.* [1991] among others.

An alternative interpretation to large magnitude rotation involves widespread partial remagnetization of cratonic sedimentary sequences [*Pan and Symon*, 1993]. Compiling late Paleozoic and early Mesozoic poles for northeastern North America, the craton interior, and the Colorado plateau, *Pan and Symons* [1993] concluded that mean poles derived from rocks in the plateau and the craton interior generally fall west (on a younger segment of the APWP) of results from northeastern North America and also that VGP distributions are "streaked" along the younger trend of the APWP. For example, the Redonda and upper shale (Bull Canyon) formations in eastern New Mexico [*Reeve and Helsley*, 1972; *Bazard and Butler*, 1991] give poles closer to the J-1 cusp than similar age poles in the Newark Supergroup and could be interpreted as partially remagnetized [*Witte and Kent*, 1990; *Witte et al.*, 1991]; Late Permian poles, however, do not appear to be affected by remagnetization [*Molina-Garza et al.*, 1989]. Partial remagnetization near the time of the J-1 cusp therefore can explain different amounts and rates of APW inferred from Chinle Group strata, particularly off the Colorado plateau where rotation is not suspected. The evidence for the interpretation of *Pan and Symons* [1993] is, however, ambiguous.

The reference pole of North America for the late Carnian-early Norian falls at about 53°N-95°E (Table 1). The mean includes data for the Triassic rift basins in eastern North America, two small intrusions in coastal Maine, continental strata in Wyoming and eastern New Mexico, and the impact structure of Manicouagan, Canada. For those poles, only the upper shale member of the Chinle Group and the Manicouagan structure lie on the craton; both yield poles to the west of other reference poles. Other poles are included in the compilation assuming tectonic stability with respect to the craton. To clarify the Triassic-Early Jurassic pole positions for North America, we have obtained paleomagnetic data from the undeformed upper Carnian Tecovas and lower Norian Trujillo Formations of the Dockum Group, northwest Texas.

Geology, Sampling, and Laboratory Techniques

The Dockum Group is of Late Triassic age [*Hunt and Lucas*, 1991]. At the base of the Dockum Group the Camp Springs Member (a thin basal conglomerate) rests disconformably on Upper Permian strata of the Quartermaster (Dewey Lake) Formation. The Tecovas Formation rests on the Camp Springs Member and is in turn disconformably overlain by the lower Norian Trujillo Formation, the youngest unit of the Dockum Group. The late Carnian age of the Tecovas Formation is based on vertebrate fauna, specifically, on the occurrence of the primitive phytosaur *Paleorhinus* in the basal conglomerate [*Hunt and Lucas*, 1991; *Lucas et al.*, 1993]. The late Carnian-early Norian age of the Dockum Group is also supported by fossil plants, restricted to the Dinophyton floral zone [*Ash*, 1980] and palynomorphs belonging to the upper Carnian New Oxford-Lockatong palynozone of the Newark Supergroup [*Cornet*, 1993].

We sampled the Dockum Group in roadcuts along Palo Duro Canyon in the Texas panhandle (Figure 2). Sites UPD1-UPD14 (Lat.=35.9°N, Long.=101.7°W; upper Palo Duro canyon) were collected in the Tecovas Formation from a 40-m-thick section in Palo Duro Canyon State Park. Along Texas State highway 284, we collected 12 additional sites in the Tecovas Formation and seven in the overlying Trujillo Formation (sites LPD1-LPD19, Lat.=34.8°N, Long=101.43°W; lower Palo Duro canyon, LPD). Each site was collected from a single stratigraphic horizon. Generalized stratigraphic sections showing the distribution of sites are depicted in Figure 2.

The Tecovas Formation consists of red mudstone with some layers and lenses of fine to coarse-grained grayish white sandstone typically occurring in thick (~0.3-1 m) beds. Sediments of the Tecovas Formation were deposited in lacustrine, fluvial, and deltaic environments. The Trujillo Formation consists of cross-

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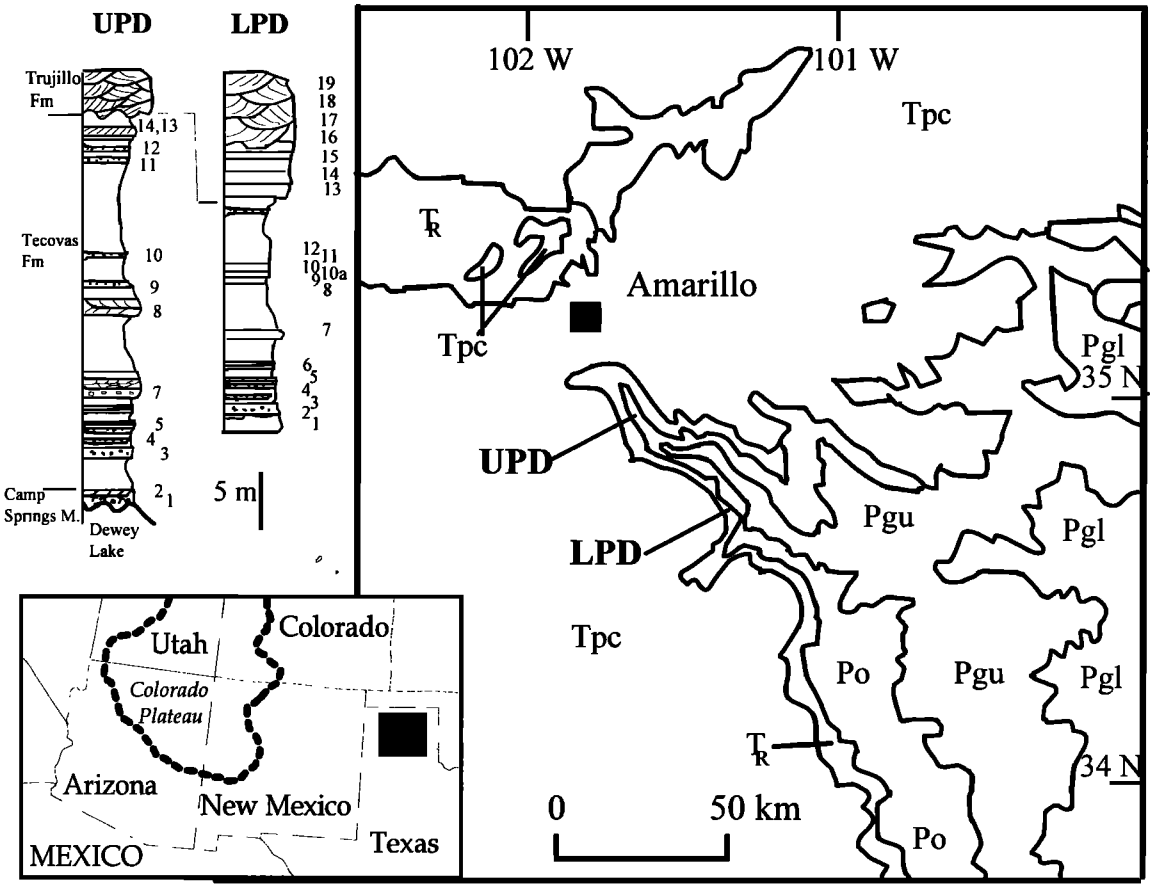


Figure 2. Generalized geologic map of northwest Texas with sampling localities (UPD, upper Palo Duro Canyon; LPD, lower Palo Duro Canyon). Sampling levels (sites) are indicated in the adjacent schematic stratigraphic sections. Abbreviations are Q, Quaternary; Tpc, Tertiary; Tr, Triassic, Po, Ochoan; Pgu, upper Guadalupian; Pgl, lower Guadalupian.

bedded, thick, fluvial, conglomeratic sandstone beds [*Johns and Granata*, 1986]. Hematite pigment is sparse in sandstone beds, which are weakly quartz cemented. Siltstone and claystone beds contain abundant hematite pigment and variable amounts of carbonate cement. At sites UPD1 and LPD2 in the Camp Springs Member we collected fine-grained pebble conglomerate (clast size ~4 mm in diameter) for a field test. We did not succeed, however, in isolating individual oriented clasts for a conglomerate test.

Samples were drilled in the field and oriented in place using a clinometer and magnetic and sun compasses. In the laboratory one or more specimens were prepared from each sample (cylinders 2.5 cm in diameter and 2.1 cm in height). Measurements of the natural remanent magnetization (NRM) were performed with either a 2G Enterprises superconducting magnetometer at the University of New Mexico or a ScT cryogenic magnetometer at the University of Michigan. Samples were subjected to progressive thermal demagnetization (7-20 steps) or alternating field (AF) demagnetization up to 140 mT, using Schonstedt or 2G instrumentation. Demagnetization results were interpreted using conventional principal component analysis [*Kirschvink*, 1980]. Site means were calculated assuming Fisherian distributions. The site mean directions and VGP distributions were analyzed with both Fisher and Bingham statistics. The results reported here are based on thermal and alternating field demagnetization of 206 samples.

Rock magnetic experiments included isothermal remanent magnetization (IRM) acquisition and demagnetization experiments (both thermal and AF). IRMs were induced using a pulse magnetizer capable of reaching inductions of 3 T, and measurements were carried out using a Schonstedt spinner magnetometer. Measurements of hysteresis parameters and low-temperature behavior of the IRM were conducted using a Micromag model 2900 at the Institute of Rock Magnetism, University of Minnesota.

Paleomagnetic Results

NRM intensities range between 0.5 and 25 mA/m. The higher intensities are typical of red siltstones and claystones of the Tecovas Formation at the LPD locality, whereas the lower intensities are typical of coarse-grained sandstones. NRM directions at UPD are scattered over the lower hemisphere, with many NRM values plotting near the recent axial-dipole direction. At LPD,

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NRM directions have south directed magnetizations streaked toward the recent axial dipole direction. Typical examples of orthogonal demagnetization diagrams [*Zijderveld*, 1967] are illustrated in Figures 3 and 4.

Samples of the Tecovas Formation exhibit two types of demagnetization behavior, which are generally related to the presence or absence of hematite pigment. For most nonhematitic specimens (devoid of hematite pigment) the NRM is characterized by one or two magnetization components (Figure 3). The quality of demagnetization diagrams in many of these specimens is marginal but does not preclude isolation of a low to intermediate laboratory unblocking temperature viscous overprint in the direction of the recent dipole field and an intermediate to high unblocking temperature characteristic magnetization (ChRM) with a good degree of confidence (MAD, maximum angular deviations of ~10° or less).

The ChRM is different from the present dipole field; it is directed to the northwest (Figures 3b, 3g, and 3h) or the southeast

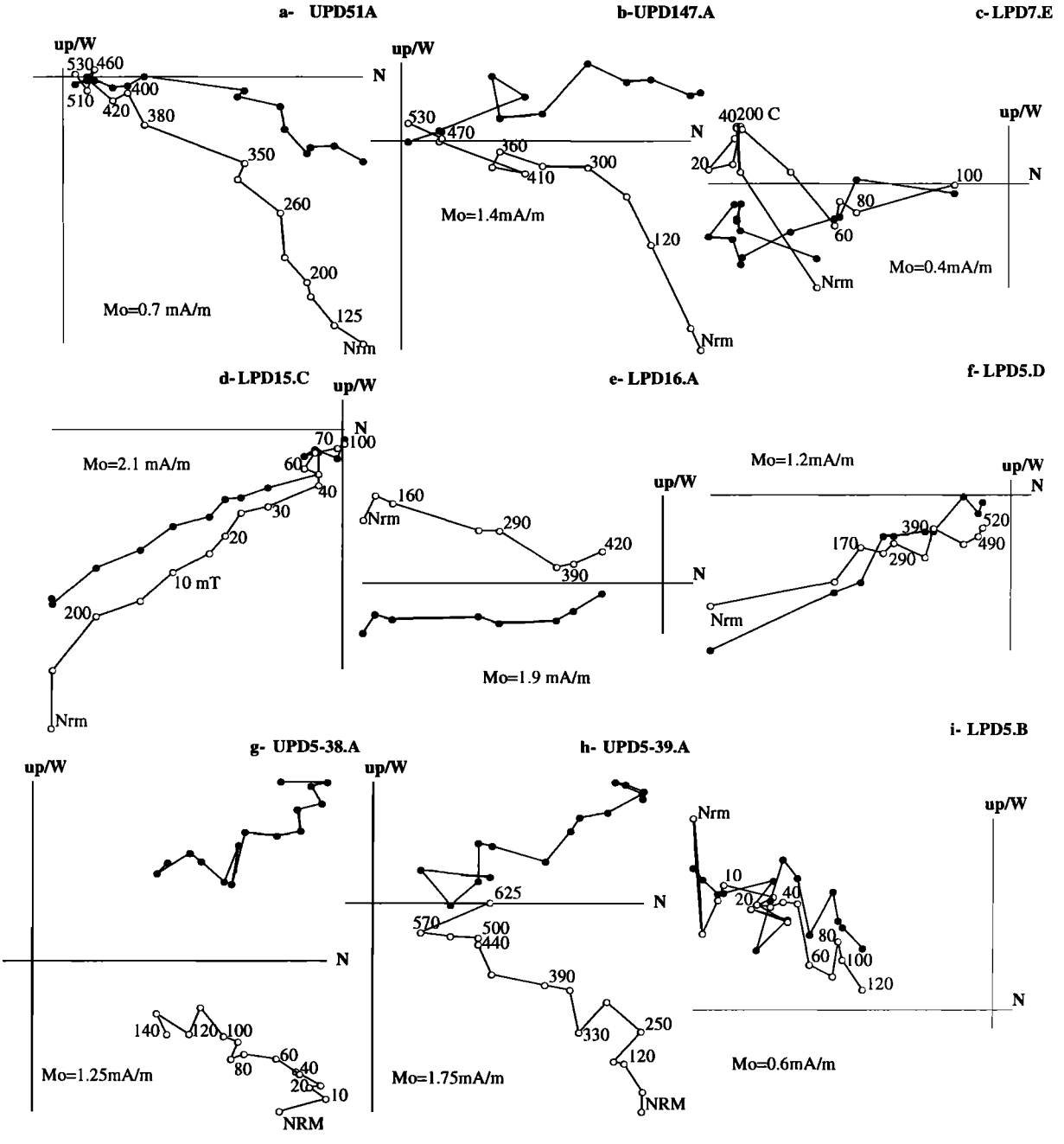


Figure 3. Orthogonal demagnetization diagrams [*Zijderveld*, 1967] of nonhematitic specimens of the Tecovas and Trujillo Formations. Open (closed) symbols indicate projections on the vertical (horizontal) plane, all specimens are plotted in field coordinates. Temperatures are in degrees Celsius (Figures 3a, 3b, 3e, 3f, and 3h) and inductions in milliTeslas (Figures 3c, 3d, 3g, 3i). Mo is the natural remanence initial intensity.

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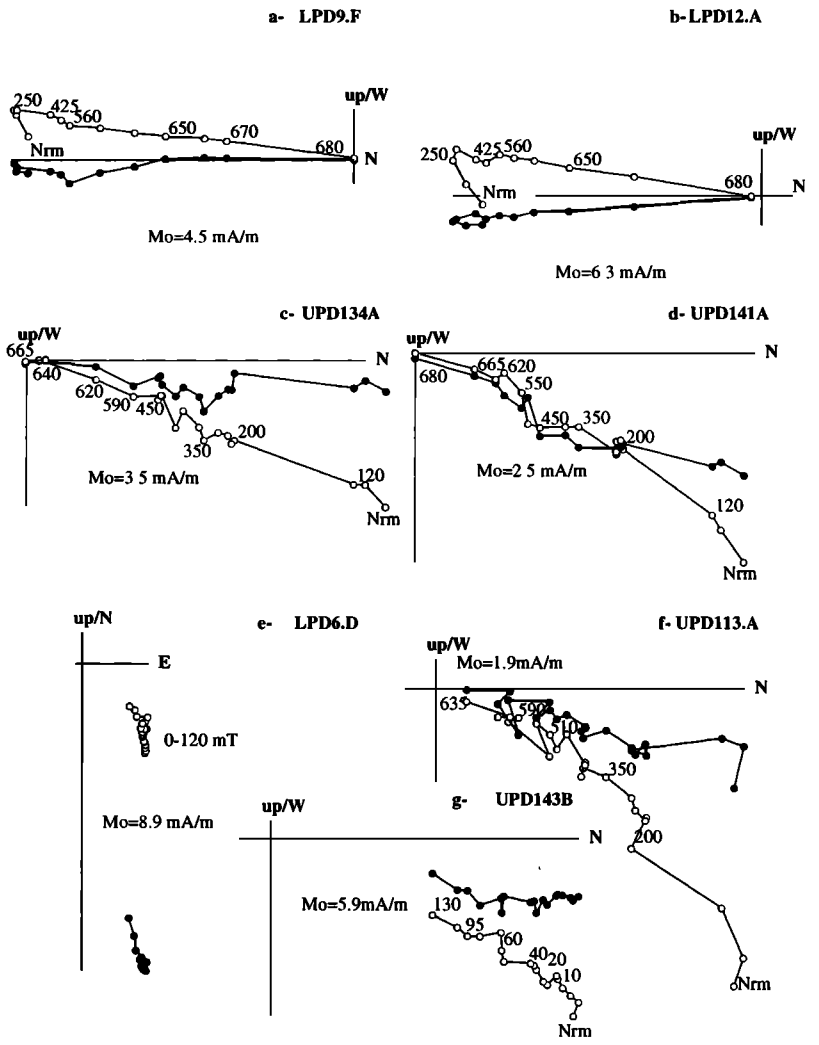


Figure 4. Orthogonal demagnetization diagrams [*Zijderveld*, 1967] of hematitic specimens of the Tecovas Formation. Symbols are as in Figure 3. Specimens in Figures 4e and 4g are examples of alternating field demagnetization results. Specimens in Figures 4d, 4f, and 4g are grayish fine-grained sandstones whose characteristic magnetization is interpreted to be dominantly carried by hematite.

with shallow to moderate positive or negative inclinations (Figures 3c, 3d, 3e, and 3f). For a number of nonhematitic specimens, thermal and AF demagnetization did not allow us to isolate an ancient magnetization (about 35% of the entire collection). Some did not respond to demagnetization, producing uninterpretable results. Others have steep and north directed magnetizations superimposed on a lower intensity, shallow magnetization component that resembles the ChRM but whose direction is poorly defined (for example Figure 3a; MAD values are greater than about 15°). Specimens exhibiting this behavior are considered unreliable, and we interpret the steep magnetization as a recent (viscous?) overprint. This magnetization is typically removed below 350°C, but gradual shallowing in inclination up to about 450°C suggests that its unblocking temperature range may be higher (Figures 3a and 3h).

Maximum temperatures of 500 to 570°C are sufficient to demagnetize samples completely, although behavior is often erratic above 500°C (Figure 3h). In nonhematitic specimens, a shallow north directed magnetization can also be isolated using AF demagnetization (e.g., Figure 3g), although coercive forces are high and anhysteretic remanence often builds up at high inductions (Figures 3g and 3i). Overlapping coercivity spectra may also preclude separation of remanence components (Figure 3d). Although laboratory unblocking temperatures do not exceed 570°C, median destructive fields are high (~60 mT). A contribution from a high-coercivity phase to the NRM is likely. In some specimens a small fraction of the NRM persists after AF inductions of 120 mT (Figures 3g and 3i). The ChRM is of reverse polarity at all the sites collected in the LPD section, whereas all sites collected at the UPD section show normal polarity magnetizations.

Red to tan (hematitic) claystones, siltstones, and sandstones have a different behavior characterized by well-defined, near single-component magnetizations (Figure 4). MAD values are

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typically less than 5°, and temperatures of up to 680°C are required to remove the NRM (Figures 4a-4d). Unblocking temperatures are distributed, with more than half of the original remanence remaining above the Curie temperature of magnetite. In hematitic specimens, low-temperature components with north and steep positive directions are small and unblocked by heating to ~350° (Figures 4b and 4d). AF demagnetization on these samples removes only a small fraction of the NRM; maximum coercivities exceed 140 mT (Figures 4e and 4g). The magnetization remaining after AF demagnetization is not appreciably different from the high unblocking temperature magnetization. Sites with gray to grayish red colored samples exhibit a characteristic magnetization which unblocks at temperatures that exceed 635°C, but AF demagnetization removes a relatively large fraction of the NRM (Figures 4f and 4g). The component removed with AF demagnetization is similar to the low unblocking temperature overprint. Demagnetization characteristics suggest that both magnetite and hematite contribute to the NRM. The ChRM was successfully isolated in at least three samples from each of 26 sites. Paleomagnetic data and statistical parameters are summarized in Table 2.

Rock Magnetism

Maximum laboratory unblocking temperatures of about 570°C and AF demagnetization response suggest that the natural remanence of nonhematitic (white) sandstones is dominated by a cubic phase (magnetite or maghemite). A minor contribution from hematite cannot be ruled out because some specimens are also characterized by high coercivities (Figure 3g). In contrast, maximum laboratory unblocking temperatures above 600° and high coercivities indicate that hematite is the most important remanence carrier in hematitic rocks. Distributed unblocking temperatures suggest that the contribution from fine-grained hematite is important.

The combination of ancient magnetizations carried by either magnetite or hematite within a sedimentary sequence only a few tens of meters in thickness is not typical of other Triassic strata in the North American southwest, although both minerals are commonly present. Both magnetite and hematite may be of detrital origin. It has been argued, however, that a large fraction of the hematite in red beds, as both coarse grains as well as pigment, is secondary [*Larson and Walker*, 1982] and that characteristic magnetizations of red beds are most typically early, but of chemical origin [*Butler*, 1992].

Isothermal remanent magnetization acquisition, progressive AF demagnetization of saturation remanent magnetization, and thermal demagnetization of two orthogonal IRM components (acquired in 1.35 T and 0.35 T) were used to further characterize the magnetic mineralogy. The IRM acquisition curves of nonhematitic (white) sandstones are nearly saturated with inductions of about 0.3 T (Figure 5a). We observed only a small (~5%) increase in magnetization between 0.3 T and the maximum induction applied (3 T). The magnetic mineralogy of these samples is clearly dominated by a cubic phase such as magnetite or maghemite, with a high-coercivity phase (or phases?) partially contributing to the IRM.

Thermal demagnetization of the IRM component induced with 1.35 T to nonhematitic samples suggests the presence of small amounts of hematite and possibly goethite. The intensity decays rapidly upon heating to 100°C and about 5% of the magnetization remains after heating to 590°C (Figure 5c). The weathered surface of these rocks is a thin (few millimeter) dark red, oxidized crust. It is likely that at least some of the high-coercivity IRM is carried by hematite or goethite grains formed in recent weathering. However, the fact that in most cases 80 to 100% of the NRM is removed with AF suggests that magnetite is the dominant carrier of the ChRM in nonhematitic strata.

Samples from red bed sites are characterized by IRM acquisition curves typical of hematite (Figure 5a). The presence of magnetite or maghemite is indicated by AF demagnetization, which removes a small part of the IRM (Figure 5b). The magnetization removed is similar to the fraction of the NRM removed by the same method. As mentioned above, thermal and AF demagnetization suggest that both magnetite and hematite may contribute to the NRM in some hematitic specimens. At temperatures below about 300°C, the intensities of both the NRM and the IRM component induced with 0.3 T decay rapidly, whereas above 400°C the unblocking temperature spectrum of the 1.35 T IRM resembles that of the ChRM (Figure 5d). There is no inflection on the intensity decay curves around the Curie temperature of magnetite. Furthermore, the ChRM exhibits coercivities in excess of 140 mT. It thus appears that a "soft" phase such as magnetite is present. It does not contribute significantly to the characteristic magnetization, but it is likely the carrier of the low-temperature overprint.

Hysteresis loops for samples of the Trujillo and Tecovas Formations are of three distinct forms (Figure 5). Red siltstones and sandstones are characterized by moderately high coercivities, greater than about 50 mT (Figure 5e). Hysteresis loops are not significantly constricted. White sandstones whose magnetization unblocks below 570° are characterized by coercivities below about 15 mT and relatively high Hcr/Hc ratios but moderate Mrs/Ms ratios. Hysteresis loops of those samples are also not constricted (Figure 5g). The third group of samples is light-colored sandstones from the UPD section whose magnetization unblocks at temperatures above 635°C. Coercivities of these samples are consistently higher than white sandstones with lower unblocking temperatures. Hysteresis loops are constricted, and Mrs/Ms ratios are relatively high (Figure 5f). The contribution of both a hard and a soft magnetic phase in these rocks seems indisputable. A logarithmic plot of hysteresis parameters (Figure 5h) shows that rocks with unblocking temperatures below 600°C have consistently higher Hcr/Hc ratios and lower Mrs/Ms ratios, suggesting that contributions from multidomain (MD) grains are important. MD grains may carry the magnetizations we interpret as recent viscous overprints.

Reflected light microscopy indicates a marked contrast in oxide mineralogy and textures between the nonhematitic and red (hematitic) rocks in both sections. In nonhematitic sandstones, visible iron oxides (mostly specular-hematite and magnetite) appear as angular to slightly embayed and fractured detrital grains (a few to tens of microns in size). Magnetite grains lack exsolution textures and hematite grains are homogeneous. Fine-grained hematite pigment in white sandstones is rare. Detrital iron oxides in rocks from the red bed sites show more advanced stages of corrosion and/or oxidation, including dissolution textures (corrosion along fractures and at grain margins, dissolution pits), skeletal trellis textures, and recrystallization. Fine-grained hematite pigment occurs as grain coatings and intergranular cement.

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Table 2. Paleomagnetic Data and Statistical Parameters, Dockum Group, Northwest Texas

Site		nd\n	Dec, deg	Inc, deg	*k*	α95, deg	VGP °N	VGP °E
Upper Palo Duro Canyon, 34.9°N-101.7°W								
Tecovas Formation								
UPD1*		7\0	...	...				
UPD2*		6\0	...	...				
UPD3	m	11\7	352.7	1.3	11.8	18.4	55.0	91.1
UPD4	m	12\4	343.4	18.1	93.4	9.6	60.2	112.8
UPD5	m	16\7	357.6	13.8	5.8	27.3	62.0	82.4
UPD7	m	13\3	347.9	12.7	9.1	43.6	59.4	102.4
UPD8	h	10\4	0.1	9.5	24.6	18.9	59.8	78.1
UPD9	h	10\7	7.7	5.9	21.4	13.3	57.2	64.0
UPD10	h	4\4	11.7	20.1	13.3	26.1	63.2	52.0
UPD11	h	11\8	4.9	7.3	13.6	17.0	58.4	68.4
UPD12	h	7\4	20.8	3.7	25.3	18.6	51.7	43.4
UPD13	h	6\6	13.8	24.3	42.9	10.3	64.5	45.6
UPD14	h	7\5	28.0	12.9	163.6	6.0	51.6	29.6
Mean		11			33.1	8.1	60.9	69.1
Lower Palo Duro Canyon, 34.7°N-101.4°W								
Tecovas Formation								
LPD1*		6\0	...	...				
LPD2*		0\0	...	...				
LPD3*		3\0	...	...				
LPD4*		2\0	...	...				
LPD5	m	5\4	184.1	-0.8	13.4	26.1	55.4	71.3
LPD6	h	4\4	174.3	1.1	470.0	4.2	54.2	88.4
LPD7	m	4\4	178.6	-2.8	7.6	35.5	56.6	81.1
LPD8	h	6\6	178.1	-9.2	92.4	7.0	59.8	82.3
LPD9	h	5\5	179.4	-7.0	342.7	4.1	58.7	79.7
LPD10a	h	5\5	188.5	-14.4	211.1	5.3	61.4	60.7
LPD10b	h	4\4	185.5	-10.2	438.1	4.4	59.9	67.6
LPD11	h	3\3	181.5	-9.0	476.8	5.7	59.7	75.0
LPD12	h	4\4	175.6	-7.9	340.1	5.0	58.9	87.1
Trujillo Formation								
LPD13*		3\0	...	...				
LPD14	m	5\4	190.0	4.6	15.2	24.3	51.7	62.3
LPD15	m	6\4	162.0	8.6	23.9	19.2	47.4	105.6
LPD16	m	5\4	155.6	-9.3	11.1	28.8	52.4	121.0
LPD17	m	6\5	163.2	2.5	8.2	28.5	50.7	105.7
LPD18	m	4\4	174.9	-14.2	46.2	13.7	62.0	89.4
LPD19	m	6\4	152.4	-1.0	10.9	29.1	47.1	121.5
Mean		15			45.1	5.8	57.2	87.9
Overall		26			35.1	4.9	59.1	80.5
UPD "A".		4	350.5	11.5	75.8	10.6	59.6	97.2
LPD "A".		8	170.1	-1.6	27.2	10.8	54.8	95.9
Combined A		12			44.2	6.6	56.4	96.3
UPD "B-n"		7	12.4	12.1	45.1	9.1	59.0	53.8
LPD "B-r"		7	180.4	-8.1	137.4	5.2	59.3	77.8
Combined B		14			63.3	5.0	59.7	65.7

Here nd\n are number of samples demagnetized\included in calculations, and k and α95 are Fisher's statistical parameters. Dec and Inc are site mean declination and inclination. Labels m and h indicate magnetite or hematite, respectively, as the dominant carrier of the characteristic magnetization. VGPs are virtual geomagnetic poles. Mean poles and distribution parameters, K and A95, where calculated using VGPs. Bingham statistics for the VGP distributions are:
All combined 59.2°N-80.6°E (*N*=26; k1=-124.8, k2=-10.9; α13=1.8°, α12=6.2°)
A combined 56.5°N-96.3°E (*N*=12; k1=-102.6, k2=-14.6; α13=2.9°, α12=7.8°)
B combined 59.7°N-65.9°E (*N*=14; k1=-204.0, k2=-19.7; α13=1.9°, α12=6.1°)
* These sites produced uninterpretable results.

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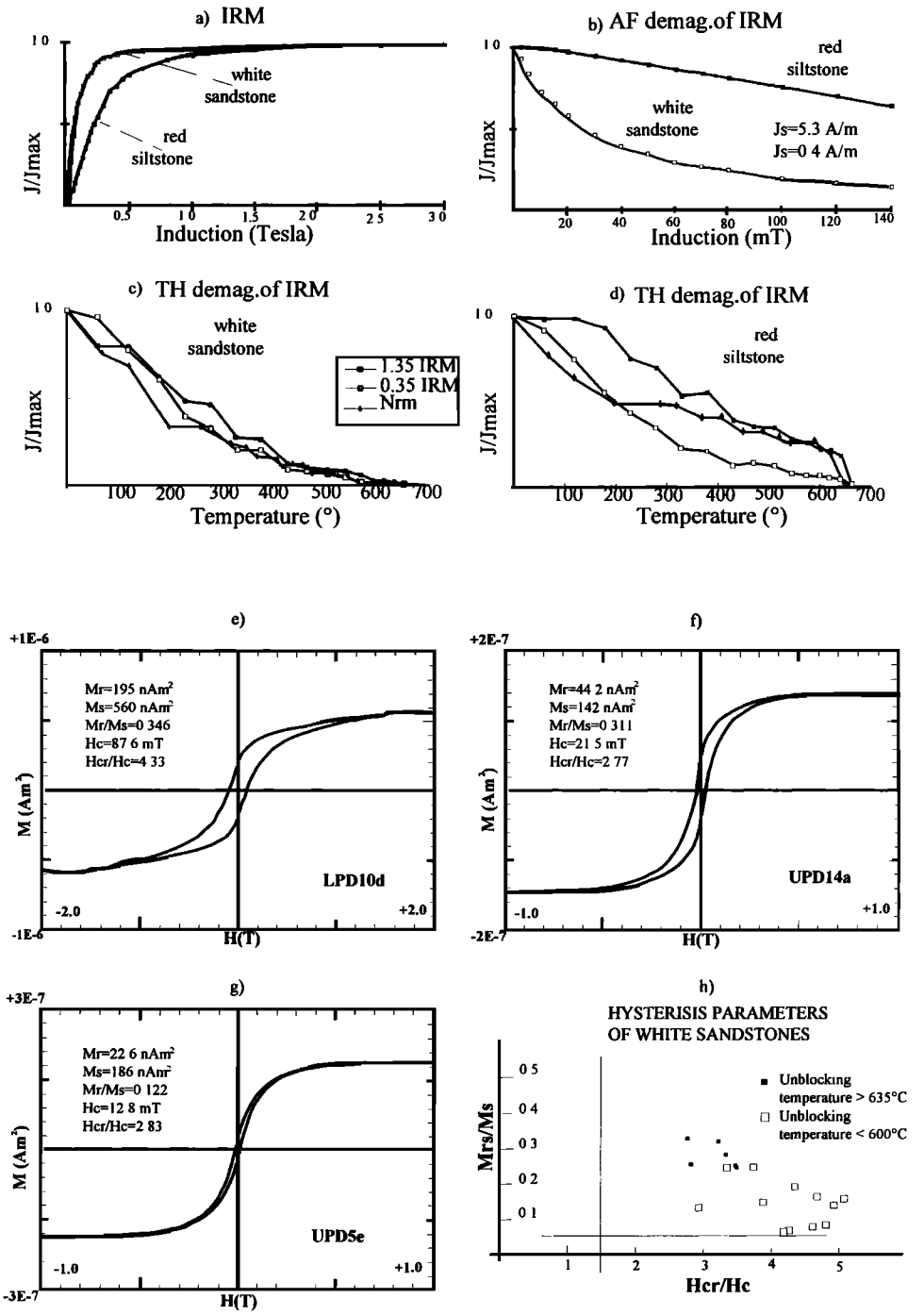


Figure 5. Rock magnetic data for selected specimens of the Tecovas Formation: (a) isothermal remanent magnetization (IRM) acquisition; (b) AF demagnetization of the remanent saturation magnetization; (c-d) thermal demagnetization of two orthogonal IRM components (induced at 0.35 and 1.35 T, respectively); (e-g) hysteresis curves and hysteresis parameters for selected specimens of the Tecovas Formation; (h) log-log plot of hysteresis parameters

Coarse (tens of microns) martite grains occur in heavy mineral laminations and in clay-rich layers stained by red pigment. Featureless specular-hematite grains are also present and compose as much as 30% of the visible iron oxide population.

Interpretation

The ensemble of 26 accepted site means yields a paleopole at 59°N-80.5°E (Table 2). This pole position is inconsistent with

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the late Carnian-early Norian reference at about 53°N-95°E. Instead, it falls on a younger segment of the North American APWP. The distribution of 26 virtual geomagnetic poles (VGPs) is "streaked" or elliptical with the major semiaxis oriented nearly east-west. This noncircular distribution of VGPs results from a large spread in declinations (Figure 6), with nonhematitic sandstones of the Tecovas and Trujillo Formations having consistently more northwest (or southeast) declinations than red bed sites.

Elliptical distributions of VGPs have been obtained from other Triassic strata in southwest North America [*Bazard and Butler*, 1991; *Molina-Garza et al.*, 1991], but a case where the VGP distribution is bimodal has not been reported. The mean pole for the UPD section is statistically different from the mean LPD pole (Table 2). The angular distance between the two poles is about 10°. Within each locality the VGP distributions remain elliptical and bimodal, with poles derived from sites whose characteristic magnetization is dominated by hematite falling along the younger segment of the APWP (Figure 1c).

When the data are divided in two groups nonhematitic (white) sites whose characteristic magnetization is dominantly carried by magnetite and hematitic (red to grayish red) sites whose characteristic magnetization is dominantly carried by hematite, the averages of site mean directions are statistically distinct at a 1% significance level. The first group yields a mean pole (pole A in Figure 1c) at 56.4°N-96.3°E (N=12, K=44.2, A95=6.6°; Table 2). The second group mean pole (B-combined) falls at 59.7°N-65.7°E (N=14, K=63.3, A95=5.0°; Table 2). The sites combined in the calculation of pole A are stratigraphically above and below the red bed sites. These include six below the red beds within the Tecovas Formation and six reverse polarity sites in the overlying Trujillo Formation. We interpret the disparity of results between the nonhematitic and red bed sites to reflect APW and therefore a significant difference in the origin and hence age of the remanence.

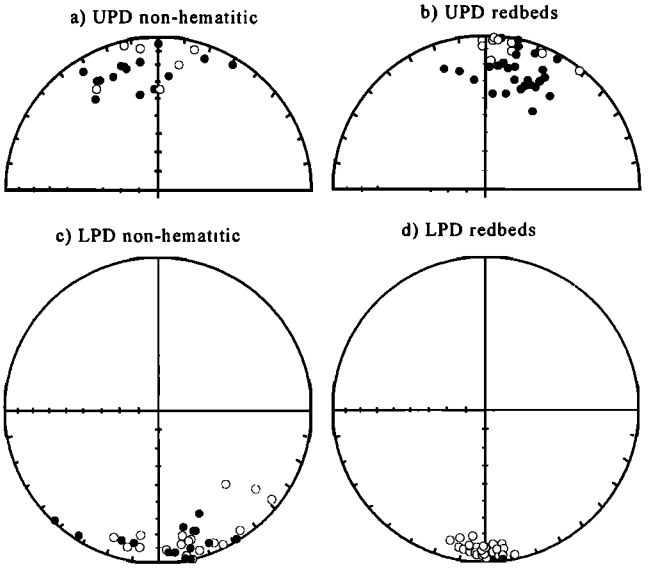


Figure 6. Equal area stereographic projections of the ChRM directions: (a-b) upper Palo Duro canyon, (c-d) lower Palo Duro canyon.

The magnetizations observed in the nonhematitic sandstones pass a reversal test (the angle between normal and reverse polarity sites is 9.9°, Table 2). The discrepancy between normal and reverse magnetizations can be explained by a small bias produced by a modern overprint. Most of the bias is eliminated by combining normal and reverse magnetizations; the mean gives a paleopole in agreement with the generally accepted late Carnian-early Norian reference pole. Magnetizations in nonhematitic sandstones could thus be interpreted as "early" magnetizations, acquired at or soon after deposition. The distribution of VGPs for nonhematitic sites exhibits some scatter, with VGPs distributed between longitudes of about 62°E and 121°E (streaked along both the younger and the older trend of the APWP). We interpret the scatter to reflect the relatively inferior fidelity of the remanence acquisition process in these rocks.

The ChRMs in the red bed sites are interpreted as secondary chemical remanent magnetizations (CRM) because they give rise to conflicting magnetostratigraphies. The red bed sections at UPD and LPD occupy similar stratigraphic positions a few meters below the base of the Trujillo sandstones but display different polarities. Furthermore, their poles fall on a younger segment of the APWP, a well-accepted criterion for suspicion of remagnetization [*Van der Voo*, 1990]. The B paleopoles derived from normal and reverse polarity magnetizations are statistically distinct at a high level of confidence (95%). The main difference between normal and reverse polarity magnetizations is in declination, suggesting that incomplete removal of a steep north directed magnetization is an unlikely explanation for the difference. We note that both subgroups, B normal and B reverse, give paleomagnetic poles along the general trend of the APWP (Figure 1c).

We suggest that separate, independent "remagnetization" events resulted in the acquisition of stratigraphically confined secondary, hematite-carried magnetizations at different times. An alternative interpretation of the paleomagnetic record reflecting continuous acquisition of a delayed CRM is unlikely.

The three most compelling arguments for discrete remagnetization events are: (1) poles A, B normal, and B reverse are statistically distinct, and B poles are derived from well grouped site means; (2) B (hematitic) sites show very low within-site dispersion (k values generally > 100, Table 2); and (3) different polarities are recorded at UPD and LPD, and we have not observed specimens carrying dual-polarity magnetizations. This suggests that the magnetization acquisition process for these rocks was effective and short-lived and argues strongly against acquisition of a long-term averaged CRM at the site level.

We also note that the stratigraphic distribution of poles is inconsistent with the expected trend of the APW path. The mean pole for the Trujillo Fm. (the 6 youngest sites) falls to the east of the B poles; if magnetizations carried by hematite in the Tecovas strata predate deposition of Trujillo strata they imply that the path back-tracked at unrealistically fast rates of APW. The rock magnetic evidence and demagnetization data are also persuasive. Rock magnetic and demagnetization data suggest that B magnetizations are dominated by (secondary?) hematite, in contrast to A magnetizations carried dominantly by magnetite or maghemite of (detrital?) origin.

Despite the small number of sites that define the B poles (14 total, 7 sites each polarity), there is no reason to believe that they represent anything but accurate records of the geomagnetic field. Within each section, red beds are of uniform magnetic polarity,

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suggesting, but not proving, acquisition during a single polarity interval. Because of the magnitude of between-site dispersion, however, it is unlikely that complete magnetization acquisition took place during a short time interval (of the order of 10^2-10^4 years) that does not average secular variation.

Origin of the Magnetization

The mechanism of magnetization acquisition in the hematitic strata of the Tecovas Formation is unclear. The saturation remanent magnetization (and the NRM) of the red beds is substantially higher than that of the nonhematitic (white) beds. Because of the large difference in saturation magnetizations between hematite and magnetite, it is unlikely that hematite in the red beds could have been derived from oxidation of detrital magnetite alone. A late acquisition of a CRM [*Larson and Walker*, 1982] does not provide a simple explanation because it does not explain the fact that possibly primary magnetizations are preserved in the nonhematitic rocks of the section.

The combined facts that paleomagnetic data for the red beds fail a reversal test and paleomagnetic poles derived from these rocks lie along a younger (latest Triassic-earliest Jurassic) segment of the APWP are clear evidence for the interpretation of the B magnetizations as secondary. A key observation is the close association between red pigment and finer-grained sediments. In fact, sandstone beds carrying B magnetizations contain thin (less than a millimeter thick) clay rich layers stained by red pigment. A possible source of Fe for hematite pigment is Fe ions originally adsorbed on clay mineral surfaces or Fe ions released from alteration of ferromagnesian silicates into clays [*Walker et al.*, 1981]. A second important observation is that the finer-grained detrital strata were originally deposited in lacustrine and marginal-lacustrine environments. We can hypothesize that sediment in such environments contained larger amounts of reduced iron detrital species and that CRM acquisition may be linked to the oxidation of such reduced species. Alternatively, complex migration paths of oxidizing fluids may also result in stratigraphic confinement of red bed sites and secondary magnetizations. As evidence to support the interpretation of a stratigraphically confined secondary magnetization, we note that Upper Permian strata of the underlying Dewey Lake Formation sampled in the same localities do not show any evidence of remagnetization [*Molina-Garza et al.*, 1989].

Discussion

The North American Apparent Polar Wander Path and the J-1 Cusp

Pole A from the Dockum Group may be combined with other North American poles of late Carnian-early Norian age to provide a refined estimate of the reference cratonic pole located at 53.0°N-94.4°E (K=210.8; A95=3.6°). In this calculation the Passaic Formation pole (np) does not include sites collected in the Jacksonwald syncline of the Newark basin because this area has experienced a small counterclockwise rotation with respect to other sites in the basin *Kodama et al.* [1994a]. In this analysis it should be noted that the poles with the younger inferred ages (upper shale, Passaic, and Manicouagan) fall consistently to the west of older poles.

Normal polarity sites at UPD from the Tecovas Formation yield a paleopole (B-n) at 59°N-54°E. We infer the age of the secondary magnetization from several lines of evidence. The pole falls to the west of the latest Norian Redonda Formation pole. The inclination of the magnetization also suggests that the age cannot be younger than about early Middle Jurassic. Furthermore, the pole is statistically indistinguishable from a pole obtained from dikes in the Piedmont province dated at 190-210 Ma, in North Carolina [*Smith*, 1987], falling at 62°N-55°E. Also, *Kodama et al.* [1994b] reported a Hettangian pole (circa 201 Ma) at 61°N-69°E from thermally remagnetized rocks of the Passaic and Bull Run Formations in the Newark and Culpeper basins, respectively. All this suggests that the B-n pole is of earliest Early Jurassic age (the approximate age of the J-1 cusp). Thus without invoking tectonic rotation in west Texas or unreasonably high rates of polar wander (with the path backtracking at similarly high velocities) the B poles can only be explained by remagnetization in latest Triassic or earliest Jurassic time.

The reverse polarity sites give a pole B-r at 60°N-78°E. Following the arguments above, the age of this remanence is bracketed between the middle Norian and late Early Jurassic (215-190 Ma); it can thus either predate or postdate the J-1 cusp of the APWP. Despite the relatively crude age control of the B poles, pole positions can be interpreted to reflect the motion of the North American plate. The B poles provide sufficient information to conclude that there was significant APW at the end of Triassic- beginning of Jurassic as previously suggested by the J-1 cusp [*Gordon et al.*, 1984; *May and Butler*, 1986].

We combined the Tecovas B poles with the Piedmont, Newark-Culpeper, and Redonda poles. The revised Late Triassic-Early Jurassic North American reference pole (J-1) falls at 60.9°N-65.6°E (A95=5.1°). The pole determination excludes the circa 200 Ma pole of the Newark Hettangian extrusive zone of *Witte and Kent* [1990] for reasons explained below. The finer detail and calibration of the APWP cannot be established from cratonic data alone because age control is poor. Paleomagnetic poles from the Colorado plateau, however, define the APWP between 215 and 190 Ma with adequate detail. The Moenave pole defines the cusp at ~200 Ma. Upper Norian-Rhaetian strata of the Church Rock Member give a pole at 59°N-63°E, (unrotated) and strata of the Pliensbachian Kayenta Formation directly overlying the Moenave Formation give a paleopole at 59°N-67°E. This indicates that the pole was already tracking eastward soon after deposition of Moenave strata [*Bazard and Butler*, 1991]. At ~210 Ma, the Church Rock pole falls on the westbound track and at ~190 Ma the Kayenta falls on the eastbound track. A position for the apex of the J-1 cusp at more westward longitudes than the Piedmont dikes, the B-n pole, and Moenave poles seems improbable. We note, however, that Early Jurassic doleritic dikes in Britanny, France, dated between 190 and 200 Ma [*Sichler and Perrin*, 1993] give a paleopole at 61°N-79°E, which rotated into North American coordinates falls between 61°N-41°E and 66°N-55°E (depending on the rotation parameters used for closure of the Atlantic ocean).

Rotation of the Colorado Plateau

The angular distance between the mean North America and Colorado plateau poles for the interval including the cusp (215-190 Ma) is only 3.5° (Figure 1d). Similar values are obtained if individual poles in that age range are compared; the angular dis-

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tances between individual poles range between 9.1° and 1.3°. We conclude that the large magnitude of rotation of the Colorado plateau inferred from Late Triassic and Early Jurassic data [e.g., *Steiner*, 1988; *Kent and Witte*, 1993] is unsubstantiated. Throughout the late Paleozoic-early Mesozoic, the Colorado plateau was an integral part of the North American plate, and there is clear evidence that rocks both on and off the plateau record polar wandering to longitudes of about 60°E.

We emphasize that the conclusion of a small plateau rotation is not solely based on the average of poles for the interval including the J-1 cusp, for which absolute ages are only crudely known. A rotation of only three degrees is sufficient to bring the mean Triassic-Jurassic Colorado plateau pole into exact agreement with the cratonic reference (using an Euler rotation pole at 36°N-105°W). This magnitude of rotation is similar but less than other estimates using Late Carboniferous (~6°), Early Permian (~4°), and late-Carnian-early Norian poles (~6°). Middle to Early Triassic poles of the Moenkopi and Chugwater Formations, however, suggest that rotation is large (~11°). Selection of a slightly different pole of rotation for the plateau may have important implications for the amount and distribution of structural deformation along the plateau boundaries [*Hamilton*, 1981, 1988] but has only negligible effect on the position of rotated paleomagnetic poles and orientations of expected directions.

A modest (maximum of 5°) to negligible amount of post-Late Triassic rotation of the Colorado plateau is compatible with geologic observations. *Kent and Witte* [1993], in proposing some 13° of clockwise rotation, skirted the critical issue of evidence in the geologic record for relatively large-magnitude plateau rotation. They failed to recognize the potential incompatibility between large-magnitude rotation and attending Late Cretaceous-early Tertiary (Laramide age) shortening and subsequent Tertiary extension, as pointed out by *Chase et al.* [1992]. Recently, *Karlstrom and Daniel* [1993] have evaluated Laramide age right-lateral strike-slip along the eastern margin of the Colorado plateau. They proposed a model that kinematically links dextral slip along the eastern margin of the plateau with coeval extension collapse of the Cordilleran orogen in central and southwest Arizona; foreland shortening in Utah, Colorado, and Wyoming; and northward translation of the plateau. The orientations of strike-slip faults they argued to have accommodated northward plateau motion are inconsistent with, as pointed out by *Chase et al.* [1992], significant clockwise rotation of the plateau. Furthermore, although not exceptionally well delineated, boundaries between Proterozoic crustal provinces and orogens, as well as the Jemez Lineament, a northeast trending crustal feature that has been active since at least Permian time and has affected sedimentary depositional patterns, crustal extension, and the distribution of igneous rocks [*Aldrich et al.*, 1986], can be traced from the Colorado plateau across its eastern margin. The orientation of these features across the eastern margin of the Colorado plateau [*Karlstrom and Daniel*, 1993] is not consistent with significant clockwise rotation of the plateau during northward translation.

A Reconciliation

A large rotation of the plateau and a very slow rate of APW were proposed to explain paleomagnetic data from the Newark Supergroup [*Kent and Witte*, 1993]. The reliability of the paleomagnetic results obtained from the Newark strata is hardly questionable. These strata contain a coherent magnetostratigraphy, pass a fold test, and contain well-defined characteristic magnetizations. Nevertheless, we have excluded the Newark Hettangian data from the cratonic average for several reasons.

The Hettangian rocks studied by *Witte and Kent* [1990] are of uniform normal polarity and no field tests could be carried out. These rocks were sampled near the border fault of the Newark basin. The border fault has been interpreted as a channel for "hot" convecting fluids. Fission-track data have been interpreted to indicate that vigorous fluid circulation in the Newark basin continued into the Middle and possibly Late Jurassic [*Steckler et al.*, 1993]. Newark strata contain widespread high-stability secondary magnetizations (interpreted to be ancient) which may partially contaminate the primary magnetization. Finally, the Newark basin may have been detached from cratonic North America, and structural complexity has been well documented along the border fault [*Kodama et al.*, 1994a]. Therefore, despite the fact that the paleomagnetic record of the Newark basin is of notable quality, the accuracy with which the Hettangian data represent cratonic plate motions is debatable.

Our data from west Texas confirm the results of *Kodama et al.* [1994b] and previously published data for Triassic strata in the Southwest, both suggesting that the cratonic reference pole for the earliest Jurassic lies near 60°N-65°E. Late acquisition of a stratigraphically confined CRM occurred in the Dockum Group; it is thus possible that remagnetization may have affected some strata in southwest North America. This form of remagnetization may have the effect of biasing Triassic pole estimates toward the J-1 cusp as proposed by *Pan and Symons* [1993] resulting in integrated, smoothed, rates of APW. But the facts remain that significant APW occurred during the Late Triassic and that estimates of the J-1 cusp on the craton and the plateau agree with one another within the statistical uncertainty. The hypotheses of a "large" rotation of the Colorado plateau and a Late Triassic near still-stand of the paleomagnetic pole between ~225 and ~200 Ma are thus inconsistent with a growing base of paleomagnetic data.

Conclusions

The poles derived from secondary magnetizations in the Tecovas Formation prompt revision of the interpretation of Triassic APW for North America. Despite being secondary magnetizations, whose ages can only be inferred by comparison with extant poles, the west Texas data confirm that the sharpness of the J-1 cusp is not an artifact of any magnitude of Colorado plateau rotation and they indicate a fast rate of APW. We have compiled a revised APW path for North America including Colorado plateau data and calculated independent estimates for the rate of apparent polar wander during Triassic time. Figure 7 shows that paleomagnetic poles from the Colorado plateau and the rest of North America both indicate similarly fast rates of apparent polar wander.

In the case of the Colorado plateau, angular distances were calculated with respect to the well-defined mean pole position of the Lower to Middle Triassic Moenkopi Formation (57.5°N - 100.2°E; N=5, A95=1.7°); whereas the Fisherian mean of previously published poles from the Chugwater Formation [*Van der Voo and Grubbs*, 1977; *Shive et al.*, 1984; *Herrera-Bervera and Helsley*, 1983] was used for the cratonic data (46.4°N° - 113.9°E; N=4, A95=5.7°). Because paleomagnetic results for

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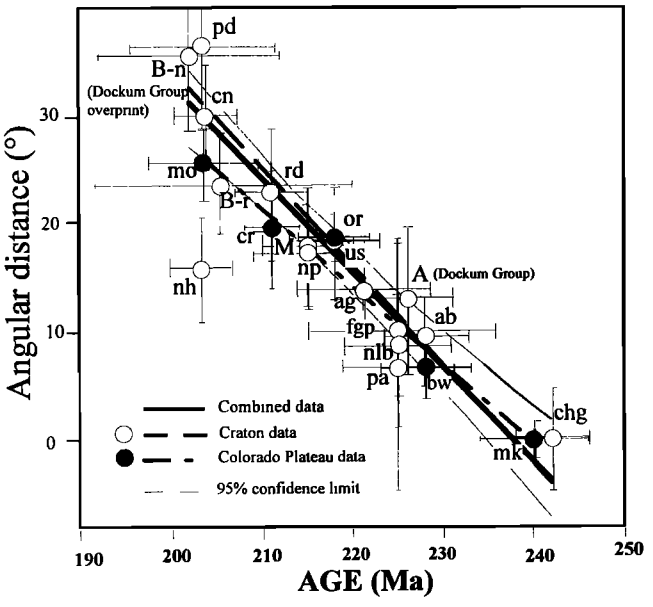


Figure 7. Average rate of APW during Triassic and earliest Jurassic time. The rate is calculated plotting the age of the pole against the angular distance to the Early-Middle Triassic (circa 240 Ma) reference. Solid symbols are used for data from the Colorado plateau. Symbols are as in Table 1. The Early-Middle Triassic reference used for the Colorado plateau is the mean of the Moenkopi Formation poles; whereas in other areas of North America, angular distances are measured with respect to the mean of the Chugwater Formation poles. Thus, estimates of the rate of APWP are independent of the rotation of the Colorado plateau. Stratigraphic ages were converted into numerical ages using the timescale of *Harland et al.* [1990]. Error bars for the angular distance correspond to a one-dimensional standard deviation (1σ); the error bars for the age indicate the stratigraphic range of the individual rock units sampled. Simple linear regression fits were computed for the Colorado plateau (dashed-dotted line), the craton (dashed line), and the combined set (solid line). Results for the plateau and the craton fall, however, within the 95% confidence region (thin dashed line) associated with the fit of the combined sets.

upper Carnian strata on the Colorado plateau are not available, we used the paleomagnetic pole for the lower Chinle Group beds (Bluewater Creek Formation) at the Joyita Hills locality in the Rio Grande valley reported by *Molina-Garza et al.* [1991]. There, the Chinle Group directly overlies Moenkopi Formation strata, and the Moenkopi Formation gives a paleopole statistically indistinguishable from paleopoles derived from this formation on the Colorado plateau in Colorado and Arizona [*Helsley and Steiner*, 1974; *Purucker et al.*, 1980; *Baag and Helsley*, 1974]. Angular distances were plotted against the respective pole age, with numerical ages assigned to stratigraphic ages using the geologic timescale of *Harland et al.* [1990]. Best estimates and error limits (1σ standard deviation) for the angular distances were calculated with an algorithm that takes into account the uncertainty in the pole position (G. Acton, personal communication, 1993). Three poles from Figure 1d were excluded from the calculations. The Kayenta Formation was excluded because it clearly postdates the cusp. The Hettangian results from the Newark Basin and the B-r pole were excluded because they may also be younger than the cusp. The results from least squares linear fits through the data in Figure 7 (craton, Colorado plateau, and combined sets) indicate at a high level of confidence (95%) that the rates of apparent polar wander suggested by data from the Colorado plateau and by other North American data do not differ. The average rate of apparent polar wander throughout the Triassic is approximately 0.85°/my (R^2=0.916). Inclusion the Newark Basin Hettangian result, which clearly falls off the general trend, only decreases the rate to 0.8°/m.y. We note the slower rate of APW suggested by data from rocks on the Colorado plateau (~0.75°/m.y.). A quadratic polynomial would provide a slightly better fit to the data (R^2=0.9308), and might suggest an increase in the rate of apparent polar wander toward the end of the Triassic and the Early Jurassic which apparently preceded the breakup of Pangea and subsequent opening of the central Atlantic. If one or more of the older poles are partially overprinted by a younger magnetization, the effect would be to "integrate" the rate of APW, thus masking more rapid velocity changes.

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